AutoVet

Inventory Forecast: The Linear Regression Equation

A simple explanation of the equation behind AutoVet's stock-out forecast, for the capstone defense.

The inventory forecast uses **simple linear regression** (scikit-learn **LinearRegression**, one input variable). The code is in **backend/ai/forecast.py** (one item) and **backend/ai/batch_forecast.py** (many items at once). Both use the same equation.

Step 1 - Prepare the data

The model does **not** fit the stock level directly, because restocks make the stock level jump up and break a straight line. Instead it fits the **running total of units used**, which only goes up (forecast.py, lines 139-145):

- **Daily usage** = how much the stock dropped that day (a restock counts as 0).
- **Running total** = daily usage added up day by day.

Step 2 - The regression equation

$$\text{Predicted } y = b_0 + b_1 x$$

Symbol	Meaning in AutoVet
x	Day number, counted from the first day of the usage history (0, 1, 2, ...)
y	Running total of units used up to that day (predicted y = the value on the fitted line)
b_1 (slope)	**Units used per day.** This is the number the forecast actually uses.
b_0 (intercept)	Where the line starts. It is saved, but not used for the prediction.

Step 3 - How b_1 and b_0 are calculated (least squares)

$$b_1 = \Sigma(x - \text{mean } x)(y - \text{mean } y) \,/\, \Sigma(x - \text{mean } x)^2$$

$$b_0 = \text{mean } y - b_1 \cdot \text{mean } x$$

Σ means "add up for every day". This is the standard line of best fit: the line with the smallest total squared error.

Step 4 - Predict when the item reaches its minimum stock

$$\text{Days left} = (\text{Current stock} - \text{Minimum stock level}) \,/\, b_1$$

The result is rounded up to a whole day (forecast.py, lines 233-234).

$$\text{Expected date} = \text{Today} + \text{Days left}$$

Result	Status shown
Fewer than 7 days left	Critical

Result	Status shown
7 to 13 days left	Reorder Soon
14 days or more	Safe
Stock already at or below the minimum	Critical
b_1 is 0 or less (item not being used)	Safe - no stockout predicted

Step 5 - How well the line fits (R^2)

$$R^2 = 1 - \Sigma(y - \text{predicted } y)^2 / \Sigma(y - \text{mean } y)^2$$

R^2	Confidence shown
0.7 or higher	High
0.4 to below 0.7	Medium
Below 0.4	Low (the system shows a warning to collect more data)

When an item has at least 10 days of data, the model is also tested properly: it learns from the first 80% of days and is scored on the last 20% (test R^2, RMSE and MAE). The forecast shown to users is then made by a model trained on 100% of the data.

Worked example

A vaccine's running total of units used gives a slope of **b_1 = 2.5 units per day**. Current stock is **40** and the minimum stock level is **10**.

Days left = (40 - 10) / 2.5 = **12 days**, so the status is **Reorder Soon** (between 7 and 13 days).

One-sentence answer for the panel

"We fit a straight line, $y = b_0 + b_1x$, to the running total of units used over time. The slope b_1 is the daily usage rate, and we divide the stock left above the minimum by that rate to predict how many days until the item needs restocking."